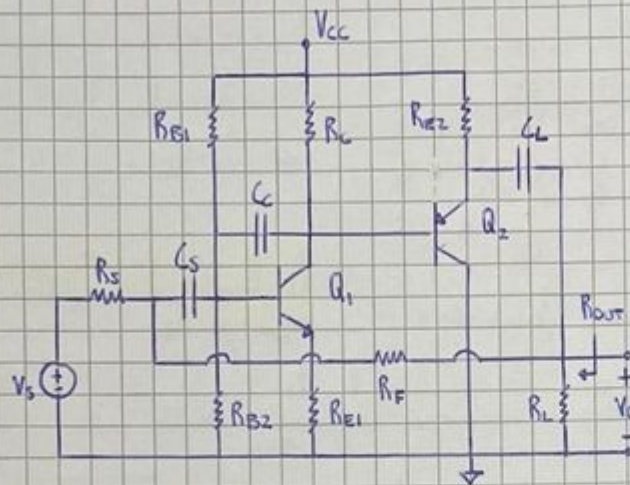


ESERCIZIO 1

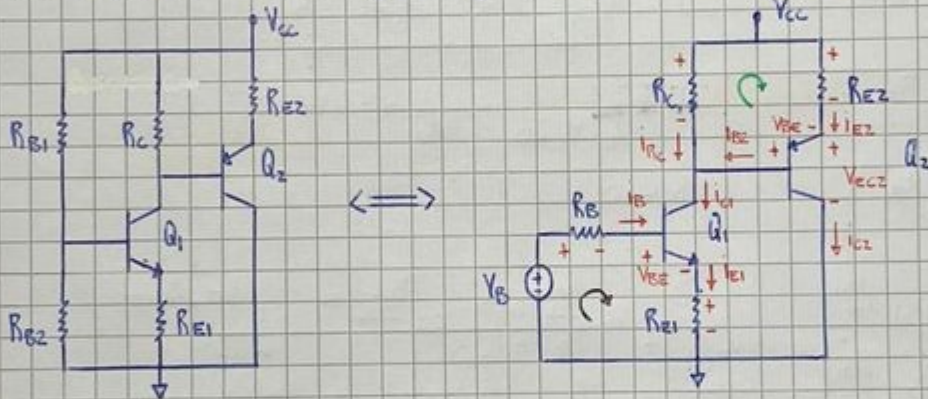


Consider the two-stage amplifier with circuit parameters, at 25°C :

$$\begin{aligned} V_{CC} &= 12\text{ V} & R_S &= 33\text{ k}\Omega \\ R_{B1} &= 120\text{ k}\Omega & R_{B2} &= 12\text{ k}\Omega \\ R_C &= 22\text{ k}\Omega & R_{E1} &= 1\text{ k}\Omega \\ R_{E2} &= 12\text{ k}\Omega & R_L &= 10\text{ k}\Omega \\ R_F &= 150\text{ k}\Omega \end{aligned}$$

$$\begin{aligned} Q_1: V_{BE} &= 0,7\text{ V} & \beta_F = \beta_0 &= 100 & r_o &\rightarrow \infty \\ Q_2: V_{BE} &= -0,7\text{ V} & \beta_F = \beta_0 &= 100 & r_o &\rightarrow \infty \\ V_T &= 25\text{ mV} \end{aligned}$$

1) DC analysis: find operating point of Q_1 and Q_2



$$V_B = V_{CC} \cdot \frac{R_{B2}}{R_{B1} + R_{B2}} = 1,05\text{ V}$$

$$R_B = R_{B1} \parallel R_{B2} = 10309\ \Omega$$

$$V_B - I_{B1} R_B - V_{BE} - (\beta_F + 1) I_{B1} R_{E1} = 0 \Rightarrow I_{B1} = \frac{V_B - V_{BE}}{R_B + (\beta_F + 1) R_{E1}} = 3,485\ \mu\text{A}$$

$$I_{C1} = \beta_F I_{B1} = 0,348\text{ mA}$$

$$I_{E1} = (\beta_F + 1) I_{B1} = 0,352\text{ mA}$$

$$R_C (I_{C1} - I_{B2}) - R_{E2} (\beta_F + 1) I_{B2} + V_{BE2} = 0 \Rightarrow I_{B2} = \frac{R_C I_{C1} + V_{BE2}}{R_C + (\beta_F + 1) R_{E2}} = 5,64\ \mu\text{A}$$

$$I_{C2} = \beta_F I_{B2} = 0,564\text{ mA}$$

$$I_{E2} = (\beta_F + 1) I_{B2} = 0,569\text{ mA}$$

$$V_{CE1} = (V_{CC} - R_C (I_{C1} - I_{B2})) - I_{E1} R_{E1} = 4,12\text{ V} > V_{CE,SAT} \rightarrow \text{OK } Q_1 \text{ in zsd}$$

$$V_{EC2} = V_{CC} - R_{E2} I_{E2} = 5,17\text{ V} > V_{CE,SAT} \rightarrow \text{OK } Q_2 \text{ in zsd}$$

$$\text{Verifica: } V_{CE1} = V_{EC2} + V_{BE2} - R_{E1} I_{E1} = 4,12\text{ V} \quad \checkmark$$

$$g_{m1} = 40 I_{C1} = 13,92\text{ mS}$$

$$r_{m1} = \beta_0 / g_{m1} = 7184\ \Omega$$

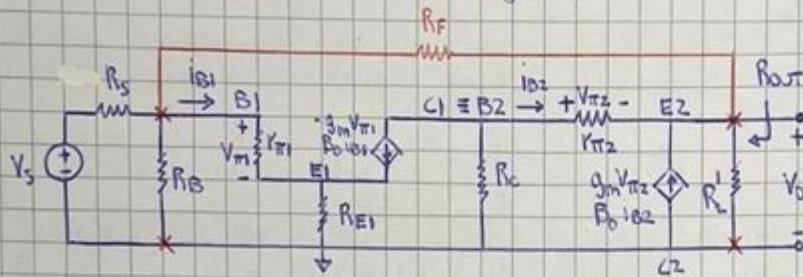
$$r_{o1} \rightarrow \infty$$

$$g_{m2} = 40 I_{C2} = 22,56\text{ mS}$$

$$r_{m2} = \beta_0 / g_{m2} = 4433\ \Omega$$

$$r_{o2} \rightarrow \infty$$

- 2) Assume C_1 and C_2 to be equivalent to short circuits and C_3 to an open circuit at the frequencies of interest. Determine the voltage gain $A_v = V_o/V_s$ of the amplifier and the output resistance R_{out} indicated in the figure.



$$R_B = R_{B1} \parallel R_{B2} = 10903 \, \Omega$$

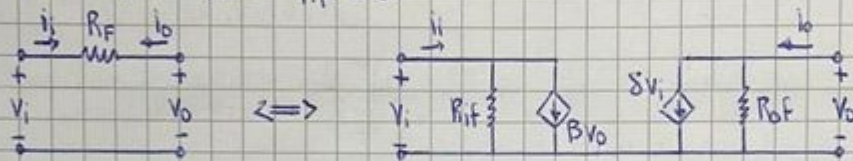
$$R_L' = R_{E2} \parallel R_L = 5455 \, \Omega$$

CE con $R_E + C_C$

su un transistor, con il feedback modifica $V_{BE} = 0$ is

Amplifier topology is trans-resistance because:

- current/shunt mixing: $i_s + i_{RF} = i_{B1}$ which drives Q_1
- voltage/shunt sampling: V_o is the sampled variable, if R_L' is shorted then $i_{RF} = 0$

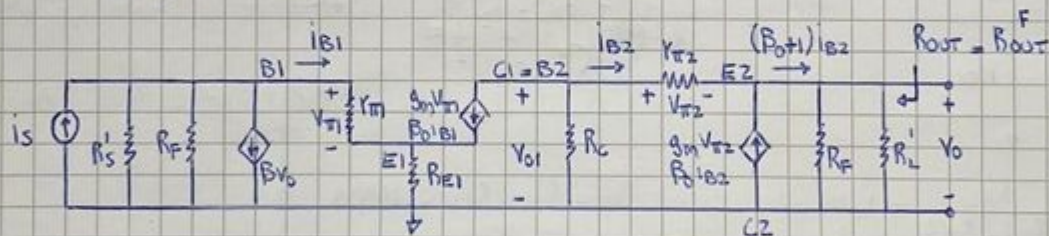


$$R_{if} \triangleq \frac{V_i}{i_i} \bigg|_{V_o=0} = R_F$$

$$\beta \triangleq \frac{i_i}{V_o} \bigg|_{V_i=0} = -\frac{1}{R_F} = -6,67 \cdot 10^{-6} \, \Omega^{-1}$$

$$R_{of} \triangleq \frac{V_o}{i_o} \bigg|_{V_i=0} = R_F$$

$$\delta \rightarrow 0$$



$$R_S' = R_S \parallel R_B = 8199 \, \Omega$$

$$i_s = V_s/R_S$$

In open loop conditions ($\beta = 0$, $\delta \rightarrow 0$):

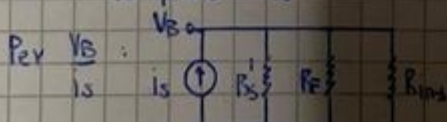
$$V_B = i_{B1} R_{\pi1} + (\beta_0 + 1) i_{B1} R_{E1} \quad V_{o1} = -\beta_0 i_{B1} R_C \parallel R_{in}^{\infty}$$

$$\Rightarrow A_{v1} \triangleq \frac{V_{o1}}{V_B} = -\frac{\beta_0 R_C \parallel R_{in}^{\infty}}{R_{\pi1} + (\beta_0 + 1) R_{E1}} = -19,53$$

$$\text{con } V_x = R_{\pi2} i_{B2} + (\beta_0 + 1) i_{B2} R_F \parallel R_L' \rightarrow R_{in}^{\infty} = R_{\pi2} + (\beta_0 + 1) R_F \parallel R_L' = 536097 \, \Omega$$

$$A_{v2} = \frac{V_o}{V_{o1}} = \frac{(\beta_0 + 1) i_{B2} R_F \parallel R_L'}{R_{\pi2} i_{B2} + (\beta_0 + 1) i_{B2} R_F \parallel R_L'} = \frac{(\beta_0 + 1) R_F \parallel R_L'}{R_{\pi2} + (\beta_0 + 1) R_F \parallel R_L'} = 0,9917$$

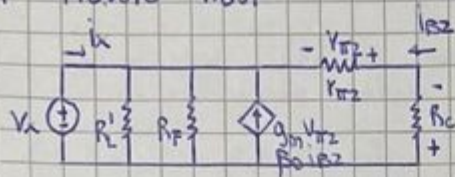
$$A_{OL} \triangleq \frac{V_o}{i_s} \bigg|_{\beta=0} = \frac{V_B}{i_s} A_{v1} A_{v2} = -1,40 \cdot 10^5 \, \Omega$$



$$V_B = i_{B1} R_{\pi1} + (\beta_0 + 1) i_{B1} R_{E1} \rightarrow R_{in1} = R_{\pi1} + (\beta_0 + 1) R_{E1} = 108184 \, \Omega$$

$$V_B/i_s = R_S' \parallel R_{in1} = 7253 \, \Omega$$

Per trovare R_{out}^{OL} :



$$V_x = -R_C i_{B2} - R_{pi} i_{B2}$$

$$i_{B2} + \beta_0 i_{B2} + i_x = 0$$

$$i_x = -(\beta_0 + 1) i_{B2}$$

$$R_{out}^{OL} = R_C \parallel R_F \parallel \frac{R_C + R_{pi}}{\beta_0 + 1} = 261,71 \, \Omega$$

In closed loop conditions:

$$A_F \triangleq \frac{V_o}{i_s} = \frac{A_{OL}}{1 + \beta A_{OL}} = -7,24 \cdot 10^4 \, \Omega \neq \frac{1}{\beta} = -1,5 \cdot 10^5 \, \Omega$$

due to low
 $T = \beta A_{OL} = 0,934$

$$A_V \triangleq \frac{V_o}{V_s} = \frac{V_o}{i_s} \cdot \frac{i_s}{V_s} = A_F \cdot K_S = \frac{A_F}{R_S} = -2,194$$

NS: β e A_{OL} hanno lo stesso segno altrimenti abbiamo fatto un feedback positivo

$$R_{out} = R_{out}^F = \frac{R_{out}^{OL}}{1 + \beta A_{OL}} = 135,32 \, \Omega$$

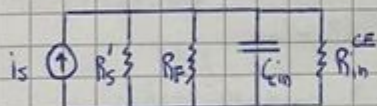
- 3) Considering $C_c = 2 \, \text{pF}$ not negligible, find f_H high frequency bandwidth limit of the amplifier (internal capacitances of Q_1 and Q_2 can still be neglected).

Il condensatore C_c è collegato tra l'ingresso e l'uscita del primo stadio: utilizzo il teorema di Miller e lo scompongo in due capacità distinte

$$C_{cin} = C_c (1 - A_{V1}) = 41 \, \text{pF}$$

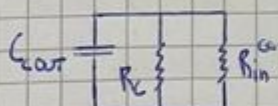
$$C_{cout} = C_c (1 - 1/A_{V1}) = 2,1 \, \text{pF}$$

Per trovare f_H applico OCTC:



$$R_{cin}^o = R_S \parallel R_F \parallel R_{in}^{CE} = 7253 \, \Omega$$

$$\text{con } R_{in}^{CE} = R_{pi} + (\beta_0 + 1) R_{E1} = 108184 \, \Omega$$



$$R_{cout}^o = R_C \parallel R_{in}^{CE} = 21133 \, \Omega$$

NS: non contare R_{in}^{CE} due volte! Già considerato sul calcolo di R_{cin}^o

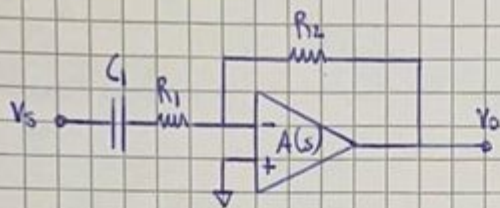
$$f_H^{OL} = \frac{\omega_H^{OL}}{2\pi} = \frac{1}{2\pi R_{cin}^o C_{cin} + R_{cout}^o C_{cout}} = 465703 \, \text{Hz}$$

Applico infine l'effetto del feedback:

$$f_H^{CL} = f_H^{OL} (1 + \beta A_{OL}) = 901 \, \text{kHz}$$

NS: sul grafico amplificazione/frequenza, se a centrobanda ho fase $0^\circ / -180^\circ$ l'amplificatore è non invertente / invertente.

ESERCIZIO 2



Consider the high-pass inverting configuration of the operational amplifier with parameters:

$$R_1 = 1 \text{ k}\Omega \quad R_2 = 100 \text{ k}\Omega \quad C_1 = 1 \text{ nF}$$

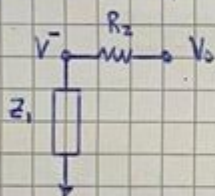
$$A(s) = 10^5 \frac{1}{1 + s/10^3} \text{ V/V}$$

- 1) After determining the expression of the ideal circuit voltage gain $A_v(s) = V_o/V_s$ and considering the given $A(s)$ expression, calculate an accurate estimation of the circuit phase margin.

È un opAmp invertente quindi:

$$A_v^{\text{IDEAL}}(s) = \frac{V_o}{V_s} = -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1 + 1/sC_1} = -\frac{sC_1 R_2}{1 + sC_1 R_1} = -C_1 R_2 \frac{s}{1 + sC_1 R_1}$$

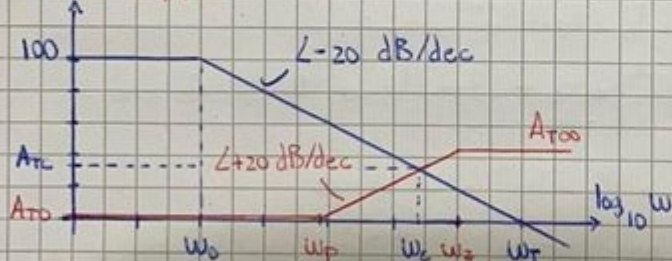
Calcolo l'espressione della rete B :



$$B \triangleq \frac{V_-}{V_o} \Big|_{V_s=0} = \frac{Z_1}{Z_1 + R_2} = \frac{1 + sC_1 R_1}{sC_1 \frac{1 + sC_1 R_1}{sC_1} + R_2} = \frac{1 + sC_1 R_1}{sC_1 \frac{1 + sC_1 R_1}{sC_1} + R_2} = \frac{1 + sC_1 R_1}{1 + sC_1 R_1 + sC_1 R_2} = \frac{1 + sC_1 R_1}{1 + sC_1 (R_1 + R_2)}$$

$$A_T \triangleq \frac{1}{B} = \frac{1 + sC_1 (R_1 + R_2)}{1 + sC_1 R_1} = A_{T0} \frac{1 + s/\omega_p}{1 + s/\omega_z}$$

$|A|_{dB}$ $|1/B|_{dB}$



$$\omega_p = \frac{1}{C_1 (R_1 + R_2)} = 9.9 \cdot 10^3 \text{ rad/s}$$

$$\omega_z = \frac{1}{R_1 C_1} = 1 \cdot 10^6 \text{ rad/s}$$

$$A_{T0} = 1 = 0 \text{ dB}$$

$$\text{CLOSING RATIO} = -20 - (+20) = -40 \text{ dB/dec}$$

$$\text{OPENING RATIO} = +20 - (-20) = +40 \text{ dB/dec}$$

$$PM \approx 0^\circ$$

$$\begin{aligned} -20 \text{ dB/dec} & \begin{cases} \omega_0 A_{T0} = \omega_c A_{Tc} \\ A_{T0}/\omega_p = A_{Tc}/\omega_c \end{cases} \sim \begin{cases} \omega_c/\omega_p = \omega_0 A_{T0} \\ A_{Tc} = \omega_c/\omega_p \end{cases} \Rightarrow \omega_c = \sqrt{\omega_0 A_{T0} \omega_p} \\ & = 3.147 \cdot 10^5 \text{ rad/s} \end{aligned}$$

$$PM = 180^\circ + \angle T(j\omega_c) = 180^\circ + \angle A(j\omega_c) + \angle B(j\omega_c) =$$

$$= 180^\circ + \angle \frac{1}{1 + j\omega_c/\omega_0} + \angle \frac{1 + j\omega_c/\omega_z}{1 + j\omega_c/\omega_p} =$$

$$= 180^\circ - \arctg\left(\frac{\omega_c}{\omega_0}\right) + \arctg\left(\frac{\omega_c}{\omega_z}\right) - \arctg\left(\frac{\omega_c}{\omega_p}\right) =$$

$$= 180^\circ - 83.98^\circ + 17.47^\circ - 88.2^\circ = 19.29^\circ$$

2) Determine the value of resistor R_1 for which the amplifier has $PM = 45^\circ$.

Per avere $PM = 45^\circ$ devo imporre la posizione dello zero a ω_c per indurre il rapporto di apertura a $+20$ dB/dec:

$$\omega_z \approx \omega_c = \frac{1}{R_1 C_1} \leadsto R_1 = \frac{1}{\omega_c C_1} = 3177,6 \, \Omega$$

Oppure, più precisamente: osservo che R_2 definisce $A_{T\infty}$:

$$A_{T\infty} = \frac{R_1 + R_2}{R_1} = 101 = 40,1 \, \text{dB}$$

Per portare il margine di fase a 45° bisogna che l'intersezione tra $A_d(s)$ e A_T non avvenga più sul tratto a $+20$ dB/dec ma sul tratto orizzontale, al limite esattamente nel punto di cambio di pendenza. I

Impongo quindi che l'intersezione avvenga a ω_c con guadagno $A_{TC} = A_{T\infty}$ e poi applico il GBP sul tratto di $A(s)$ a -20 dB/dec:

$$\omega_c A_{TC} = A_{d0} \omega_0 \quad \text{e} \quad A_{TC} = \frac{R_1 + R_2}{R_1}$$

$$\leadsto \frac{R_2}{R_1} = \frac{A_{d0} \omega_0}{\omega_c} - 1 \quad \Rightarrow \quad R_1 = \frac{R_2 \omega_c}{A_{d0} \omega_0 - \omega_c} = 3249 \, \Omega$$

3) Determine the voltage gain at $\omega = 10^7$ rad/s, taking into account the specified $A(s)$.

Osservo che $\omega = 10^7$ rad/s = ω_T , la frequenza in cui $|A_d(j\omega_T)| = 1$.

$$\text{A } \omega_T \quad |BA_{OL}| \ll 1 \quad \text{quindi} \quad A_F = \frac{A_{OL}}{1 + BA_{OL}} \times A_{OL} \Rightarrow |A_V(j\omega_T)| \approx 1$$

NB: nessuna rete di retroazione mi può dare $|A_V(s)| > |A_d(s)|$

Più precisamente:

$$A_F = W_{in} \cdot A_F = W_{in} \frac{A_{OL}}{1 + BA_{OL}} \stackrel{\omega_T}{=} W_{in} \frac{1}{1 + B}$$

Calcolo W_{in} :

$$W_{in}(s) \triangleq \frac{V_4}{V_s} \Big|_{V_o=0} = -\frac{R_2}{Z_1 + R_2} = -\frac{R_2}{\frac{1+sC_1 R_1}{sC_1} + R_2} = -\frac{sC_1 R_2}{1 + sC_1(R_1 + R_2)}$$

$$\omega_c = 10^4 \, \text{rad/s} \\ \omega_p = 9,7 \cdot 10^3 \, \text{rad/s}$$

A $\omega = \omega_T$, i valori di $B(s)$ e $W_{in}(s)$ sono costanti:

$$\Rightarrow |A_V(j\omega_T)| = |W_{in}(j\omega_T)| \cdot \frac{1}{1 + |B(j\omega_T)|} = -\frac{R_2}{R_1 + R_2} \cdot \frac{1}{1 + \frac{R_1}{R_1 + R_2}} =$$

$$= \frac{-R_2}{2R_1 + R_2} = \begin{cases} -0,94 & \text{con } R_1 = 3249 \, \Omega \\ -0,98 & \text{con } R_1 = 1000 \, \Omega \end{cases}$$